

GlobalRoute Freight Performance & Delay Diagnostic

Executive Summary:

The GlobalRoute Freight Performance & Delay Diagnostic Dashboard provides a consolidated view of supply chain operations, customer value, and freight performance. It highlights order efficiency, delivery delays, profitability, and category-level sales trends across global markets. This dashboard enables stakeholders to identify bottlenecks, monitor KPIs, and make data-driven decisions to improve logistics and customer satisfaction.

Data Sources:

- *DataCoSupplyChainDataset.csv (transactional data: orders, shipments, customers, products).*
- *DescriptionDataCoSupplyChain.csv (metadata definitions for fields such as Order Status, Shipping Mode, Late Delivery Risk, Customer Segment, etc.).*
- *tokenized_access_logs.csv (system logs for operational monitoring).*

Key fields:

Order Status: COMPLETE, PENDING, CLOSED, etc.

Mode: Standard Class, First Class, Second Class, Same Day.

Late_delivery_risk: Binary indicator (1 = late, 0 = on time).

Sales, Profit, Customer LTV, Avg Order Value: Core financial metrics.

Market & Region: Geographic segmentation (Africa, Shipping Europe, LATAM, Pacific Asia, USCA).

Project Aim:

The aim of the GlobalRoute Freight Performance & Delay Diagnostic Project is to transform raw supply chain data into actionable insights by building interactive dashboards that monitor freight performance, order management, customer value, and profitability across global markets.

This project seeks to:

- ***Provide a single source of truth for logistics and order data.***
- ***Enable stakeholders to identify inefficiencies in shipping modes and regional freight operations.***
- ***Improve customer satisfaction by reducing late deliveries and pending orders.***
- ***Support profitability and growth through analysis of discounts, margins, and product category sales.***

Project Objective

1. Operational Objectives

- **Reduce Late Deliveries:** Analyze shipping modes and regional freight data to lower the late delivery rate.
- **Optimize Shipping Modes:** Identify inefficiencies in Standard Class and encourage faster modes (First Class, Same Day).
- **Streamline Order Processing:** Reduce pending and processing orders by automating workflows and improving approvals.

2. Customer-Centric Objectives

- **Enhance Customer Satisfaction:** Increase on-time delivery percentage to improve customer loyalty.
- **Monitor Customer Lifetime Value (LTV):** Track and grow LTV by identifying high-value customers and retention strategies.
- **Reduce Cancellations & Fraud:** Minimize canceled and suspected fraud orders through better monitoring and validation.

3. Financial Objectives

- **Increase Profit Margin:** Raise profit margin by balancing discounts and freight costs.
- **Revenue Growth:** Maintain stable monthly sales while identifying opportunities for seasonal growth.
- **Discount Optimization:** Monitor total discounts and ensure promotions align with profitability goals.

4. Analytical Objectives

- **Develop KPIs & Formulas:** Define and implement metrics such as Late Delivery Rate, Avg Order Value, Profit Margin %, and Customer LTV.
- **Build Interactive Dashboards:** Create pages for Freight & Order Overview, Delivery & Category Sales, and Global Freight Performance.
- **Validate Data Accuracy:** Cross-check Power BI outputs with SQL queries and raw CSV fields.

5. Strategic Objectives

- **Market Insights:** Compare freight performance across Africa, Europe, LATAM, Pacific Asia, and USCA.
- **Product Category Trends:** Highlight top-performing categories (Computers, Cameras) and identify growth opportunities in niche categories.
- **Future Enhancements:** Prepare groundwork for predictive analytics (delivery delay forecasting) and ERP integration.

Project scope of work:

The project covers:

- **Data Preparation:** Cleaning, transformation, and integration of CSV datasets (orders, customers, products, freight).
- **Dashboard Development:** Designing three main pages — Freight & Order Overview, Delivery & Category Sales, Global Freight Performance.
- **KPI Definition:** Establishing formulas for metrics such as Late Delivery Rate, Profit Margin %, Customer LTV.
- **Insights & Recommendations:** Providing actionable strategies for logistics optimization and profitability improvement.
- **Validation & Troubleshooting:** Ensuring accuracy through SQL checks, Power BI validations, and anomaly detection.

Datasets:

DataCoSupplyChain:

A		B		C		D		E		F		G		H		I		J		K	
Type	Days for shipping (real)	Days for shipment (scheduled)	Benefit per order	Sales per customer	Delivery Status	Late delivery risk	Category Id	Category Name	Customer City	Customer Country	Custo										
DEBIT	3	4	91.25	314.6400146	Advance shipping	0	73	Sporting Goods	Caguas	Puerto Rico	XXXXX										
TRANSFER	5	4	-249.0899963	311.3599854	Late delivery	1	73	Sporting Goods	Caguas	Puerto Rico	XXXXX										
CASH	4	4	-247.7799988	309.7200012	Shipping on time	0	73	Sporting Goods	San Jose	EE. UU.	XXXXX										
DEBIT	3	4	22.8600061	304.8099976	Advance shipping	0	73	Sporting Goods	Los Angeles	EE. UU.	XXXXX										
PAYMENT	2	4	134.2100067	298.25	Advance shipping	0	73	Sporting Goods	Caguas	Puerto Rico	XXXXX										
TRANSFER	6	4	18.57999992	294.980011	Shipping canceled	0	73	Sporting Goods	Tonawanda	EE. UU.	XXXXX										
DEBIT	2	1	95.18000031	288.4200134	Late delivery	1	73	Sporting Goods	Caguas	Puerto Rico	XXXXX										
TRANSFER	2	1	68.43000031	285.1400146	Late delivery	1	73	Sporting Goods	Miami	EE. UU.	XXXXX										
CASH	3	2	133.7200012	278.5899963	Late delivery	1	73	Sporting Goods	Caguas	Puerto Rico	XXXXX										
CASH	2	1	132.1499939	275.3099976	Late delivery	1	73	Sporting Goods	San Ramon	EE. UU.	XXXXX										
TRANSFER	6	2	130.5800018	272.0299988	Shipping canceled	0	73	Sporting Goods	Caguas	Puerto Rico	XXXXX										
TRANSFER	5	2	45.6899963	268.7600098	Late delivery	1	73	Sporting Goods	Freeport	EE. UU.	XXXXX										
TRANSFER	4	2	21.76000023	262.2000122	Late delivery	1	73	Sporting Goods	Salinas	EE. UU.	XXXXX										
DEBIT	2	1	24.57999992	245.8099976	Late delivery	1	73	Sporting Goods	Caguas	Puerto Rico	XXXXX										
TRANSFER	2	1	16.38999939	327.75	Late delivery	1	73	Sporting Goods	Peabody	EE. UU.	XXXXX										
DEBIT	2	1	-259.5799866	324.4700012	Late delivery	1	73	Sporting Goods	Caguas	Puerto Rico	XXXXX										
PAYMENT	5	2	-246.3600006	321.2000122	Late delivery	1	73	Sporting Goods	Canovanas	Puerto Rico	XXXXX										
CASH	2	1	23.84000015	317.9200134	Late delivery	1	73	Sporting Goods	Paramount	EE. UU.	XXXXX										
DEBIT	2	1	102.2600021	314.6400146	Late delivery	1	73	Sporting Goods	Caguas	Puerto Rico	XXXXX										
PAYMENT	0	0	87.18000031	311.3599854	Shipping on time	0	73	Sporting Goods	Mount Prospect	EE. UU.	XXXXX										
TRANSFER	0	0	154.8600006	309.7200012	Shipping on time	0	73	Sporting Goods	Long Beach	EE. UU.	XXXXX										
TRANSFER	5	4	82.30000305	304.8099976	Late delivery	1	73	Sporting Goods	Caguas	Puerto Rico	XXXXX										
TRANSFER	4	2	22.37000084	298.25	Late delivery	1	73	Sporting Goods	Rancho Cordova	EE. UU.	XXXXX										
TRANSFER	3	2	17.70000076	294.980011	Shipping canceled	0	73	Sporting Goods	Caguas	Puerto Rico	XXXXX										
TRANSFER	2	2	90.27999978	288.4200134	Shipping canceled	0	73	Sporting Goods	Billings	EE. UU.	XXXXX										
DEBIT	6	2	131.1699982	285.1400146	Late delivery	1	73	Sporting Goods	Caguas	Puerto Rico	XXXXX										
TRANSFER	5	2	90.54000092	278.5899963	Late delivery	1	73	Sporting Goods	Wilkes Barre	EE. UU.	XXXXX										
PAYMENT	4	4	82.58999634	275.3099976	Shipping on time	0	73	Sporting Goods	Caguas	Puerto Rico	XXXXX										
DEBIT	3	4	-17.13999939	272.0299988	Advance shipping	0	73	Sporting Goods	Roseville	EE. UU.	XXXXX										
TRANSFER	2	4	23.64999962	268.7600098	Advance shipping	0	73	Sporting Goods	Bellflower	EE. UU.	XXXXX										
DEBIT	6	4	73.41999817	262.2000122	Late delivery	1	73	Sporting Goods	Caguas	Puerto Rico	XXXXX										
DEBIT	6	4	119.8000007	245.8099976	Late delivery	1	73	Sporting Goods	Mt Pleasant	EE. UU.	XXXXX										
Description: DataCoSupplyChain		Authorized access: User		DataCoSupplyChainDataset		Columns:		1		2		3		4		5		6		7	

Tokenized_access-logs:

Under Armour Women's Micro G Skulpt Running S	boxing & mma	09-01-2017 06:06	Sep	6	footwear	171.218.47.35	/department/footwear/category/boxing%20&%20mma/product/Under%20Armour%20Women's%20Micro%20G%20Skulpt%20Running%20S
Under Armour Men's Compression EV SL Slide	electronics	09-01-2017 06:06	Sep	6	footwear	194.78.95.39	/department/footwear/category/electronics/product/Under%20Armour%20Men's%20Compression%20EV%20SL%20Slide
SOLE ESS Elliptical	strength training	09-01-2017 06:07	Sep	6	footwear	41.209.131.244	/department/footwear/category/strength%20training/product/SOLE%20ESS%20Elliptical%20to%20cart
Nike Men's Dri-FIT Victory Golf Polo	women's apparel	09-01-2017 06:07	Sep	6	golf	81.232.70.160	/department/golf/category/women's%20apparel/product/Nike%20Men's%20Dri-FIT%20Victory%20Golf%20Polo
Under Armour Hustle Storm Medium Duffel Bag	fitness accessories	09-01-2017 06:08	Sep	6	footwear	131.132.236.70	/department/footwear/category/fitness%20accessories/product/Under%20Armour%20Hustle%20Storm%20Medium%20Duffel%20Bag
Nike Men's Comfort 2 Slide	tennis & racquet	09-01-2017 06:09	Sep	6	fitness	186.82.23.71	/department/fitness/category/tennis%20&%20racquet/product/Nike%20Men's%20Comfort%202%20Slide
Under Armour Men's Tech II T-Shirt	lacrosse	09-01-2017 06:09	Sep	6	fitness	7.176.104.99	/department/fitness/category/lacrosse/product/Under%20Armour%20Men's%20Tech%20II%20T-Shirt
Nike Men's Free 5.0+ Running Shoe	cardio equipment	09-01-2017 06:09	Sep	6	footwear	186.82.23.71	/department/footwear/category/cardio%20equipment/product/Nike%20Men's%20Free%205.0+%20Running%20Shoe
Nike Men's Fingertrap Max Training Shoe	soccer	09-01-2017 06:10	Sep	6	fitness	64.75.129.109	/department/fitness/category/soccer/product/Nike%20Men's%20Fingertrap%20Max%20Training%20Shoe
adidas Youth Germany Black/Red Away Match Soc	girls' apparel	09-01-2017 06:10	Sep	6	golf	187.193.85.79	/department/golf/category/girls%20apparel/product/adidas%20Youth%20Germany%20Black%20Red%20Away%20Match%20Soccer%20Shoe
Pelican Sunstream 100 Kayak	water sports	09-01-2017 06:11	Sep	6	fan shop	173.253.223.50	/department/fan%20shop/category/water%20sports/product/Pelican%20Sunstream%20100%20Kayak
Field & Stream Sportsman 16 Gun Fire Safe	fishing	09-01-2017 06:12	Sep	6	fan shop	57.209.125.46	/department/fan%20shop/category/fishing/product/Field%20&%20Stream%20Sportsman%2016%20Gun%20Fire%20Safe
Yakima DoubleDown Ace Hitch Mount 4-Bike Rack	strength training	09-01-2017 06:12	Sep	6	footwear	217.154.105.170	/department/footwear/category/strength%20training/product/Yakima%20DoubleDown%20Ace%20Hitch%20Mount%204-Bike%20Rack
Columbia Men's PFG Anchor Tough T-Shirt	shop by sport	09-01-2017 06:13	Sep	6	golf	62.164.176.196	/department/golf/category/shop%20by%20sport/product/Columbia%20Men's%20PFG%20Anchor%20Tough%20T-Shirt
Perfect Fitness Perfect Rip Deck	cleats	09-01-2017 06:13	Sep	6	apparel	26.26.6.78	/department/apparel/category/cleats/product/Perfect%20Fitness%20Perfect%20Rip%20Deck
Ogio Race Golf Shoes	golf bags & carts	09-01-2017 06:14	Sep	6	outdoors	73.239.207.127	/department/outdoors/category/golf%20bags%20&%20carts/product/Ogio%20Race%20Golf%20Shoes%20Add%20To%20Cart
Stiga Master Series ST3100 Competition Indoor	hockey	09-01-2017 06:15	Sep	6	fitness	26.26.6.78	/department/fitness/category/hockey/product/Stiga%20Master%20Series%20ST3100%20Competition%20Indoor%20Table%20Tennis%20Table
Diamondback Boys' Insight 24 Performance Hybr	basketball	09-01-2017 06:15	Sep	6	fitness	64.233.242.248	/department/fitness/category/basketball/product/Diamondback%20Boys%20Insight%2024%20Performance%20Hybrid%20Bike
adidas Youth Germany Black/Red Away Match Soc	girls' apparel	09-01-2017 06:15	Sep	6	golf	187.193.85.79	/department/golf/category/girls%20apparel/product/adidas%20Youth%20Germany%20Black%20Red%20Away%20Match%20Soccer%20Shoe
Nike Men's Cj Elite 2 TD Football Cleat	men's footwear	09-01-2017 06:15	Sep	6	apparel	210.138.158.124	/department/apparel/category/men's%20footwear/product/Nike%20Men's%20CJ%20Elite%202%20TD%20Football%20Cleat
Columbia Men's PFG Anchor Tough T-Shirt	shop by sport	09-01-2017 06:15	Sep	6	golf	64.75.129.109	/department/golf/category/shop%20by%20sport/product/Columbia%20Men's%20PFG%20Anchor%20Tough%20T-Shirt
O'Brien Men's Neoprene Life Vest	indoor outdoor games	09-01-2017 06:16	Sep	6	fan shop	52.173.216.27	/department/fan%20shop/category/indoor%20outdoor%20games/product/O'Brien%20Men's%20Neoprene%20Life%20Vest
The North Face Women's Recon Backpack	hunting & shooting	09-01-2017 06:16	Sep	6	fan shop	63.241.234.226	/department/fan%20shop/category/hunting%20&%20shooting/product/The%20North%20Face%20Women's%20Recon%20Backpack
adidas Men's Germany Black Crest Away Tee	girls' apparel	09-01-2017 06:16	Sep	6	golf	63.241.234.226	/department/golf/category/girls%20apparel/product/adidas%20Men's%20Germany%20Black%20Crest%20Away%20Tee
O'Brien Men's Neoprene Life Vest	women's apparel	09-01-2017 06:17	Sep	6	golf	118.186.177.193	/department/golf/category/women's%20apparel/product/Nike%20Men's%20Dri-FIT%20Victory%20Golf%20Polo
O'Brien Men's Neoprene Life Vest	indoor outdoor games	09-01-2017 06:17	Sep	6	fan shop	197.41.145.14	/department/fan%20shop/category/indoor%20outdoor%20games/product/O'Brien%20Men's%20Neoprene%20Life%20Vest
Diamondback Boys' Insight 24 Performance Hybr	basketball	09-01-2017 06:18	Sep	6	fitness	130.130.149.74	/department/fitness/category/basketball/product/Diamondback%20Boys%20Insight%2024%20Performance%20Hybrid%20Bike
Bag Boy Beverage Holder	golf gloves	09-01-2017 06:18	Sep	6	outdoors	197.41.145.14	/department/outdoors/category/golf%20gloves/product/Bag%20Boy%20Beverage%20Holder
O'Brien Men's Neoprene Life Vest	indoor outdoor games	09-01-2017 06:18	Sep	6	fan shop	73.239.207.127	/department/fan%20shop/category/indoor%20outdoor%20games/product/O'Brien%20Men's%20Neoprene%20Life%20Vest
O'Brien Men's Neoprene Life Vest	indoor outdoor games	09-01-2017 06:19	Sep	6	fan shop	130.130.149.74	/department/fan%20shop/category/indoor%20outdoor%20games/product/O'Brien%20Men's%20Neoprene%20Life%20Vest
Bridgestone e6 Straight Distance NFL Tennessee	electronics	09-01-2017 06:19	Sep	6	outdoors	197.41.145.14	/department/outdoors/category/electronics/product/Bridgestone%20e6%20Straight%20Distance%20NFL%20Tennessee

DescriptionDataCoSupplyChain

FIELDS	DESCRIPTION
Type	: Type of transaction made
Days for shipping (real)	: Actual shipping days of the purchased product
Days for shipment (scheduled)	: Days of scheduled delivery of the purchased product
Benefit per order	: Earnings per order placed
Sales per customer	: Total sales per customer made per customer
Delivery Status	: Delivery status of orders: Advance shipping , Late delivery , Shipping canceled , Shipping on time
Late_delivery_risk	: Categorical variable that indicates if sending is late (1), it is not late (0).
Category Id	: Product category code
Category Name	: Description of the product category
Customer City	: City where the customer made the purchase
Customer Country	: Country where the customer made the purchase
Customer Email	: Customer's email
Customer Fname	: Customer name
Customer Id	: Customer ID
Customer Lname	: Customer lastname
Customer Password	: Masked customer key
Customer Segment	: Types of Customers: Consumer , Corporate , Home Office
Customer State	: State to which the store where the purchase is registered belongs
Customer Street	: Street to which the store where the purchase is registered belongs
Customer Zipcode	: Customer Zipcode
Department Id	: Department code of store
Department Name	: Department name of store
Latitude	: Latitude corresponding to location of store
Longitude	: Longitude corresponding to location of store
Market	: Market to where the order is delivered : Africa , Europe , LATAM , Pacific Asia , USCA
Order City	: Destination city of the order
Order Country	: Destination country of the order
Order Customer Id	: Customer order code
order date (DateOrders)	: Date on which the order is made
Order Id	: Order code
Order Item Cardprod Id	: Product code generated through the RFID reader

Dim_customer table:

Customer City ▾	Customer Country ▾	Customer Fname ▾	Customer Id ▾	Customer Lname ▾	Customer Segment ▾	Customer State ▾	Customer Street ▾
Caguas	Puerto Rico	Mary	8917	Smith	Consumer	PR	4992 Jagged Forest Subdivision
Caguas	Puerto Rico	Mary	7784	Smith	Consumer	PR	3216 Red Glade
Caguas	Puerto Rico	Mary	2737	Smith	Consumer	PR	5424 Rustic Anchor Boulevard
Caguas	Puerto Rico	Mary	6428	Smith	Consumer	PR	3567 Heather Knoll
Caguas	Puerto Rico	Mary	5118	Smith	Consumer	PR	1388 Blue Turnabout
Caguas	Puerto Rico	Mary	7247	Smith	Consumer	PR	6401 Shady Embers Farms
Caguas	Puerto Rico	Mary	6887	Smith	Consumer	PR	8840 Harvest Towers
Caguas	Puerto Rico	Mary	10668	Smith	Consumer	PR	5087 Crystal Hills Manor
Caguas	Puerto Rico	Mary	8456	Smith	Consumer	PR	9198 Rocky Run
Caguas	Puerto Rico	Mary	6523	Smith	Consumer	PR	8094 Broad Dale
Caguas	Puerto Rico	Mary	4867	Smith	Consumer	PR	6292 Velvet Wagon Plaza
Caguas	Puerto Rico	Mary	4998	Smith	Consumer	PR	1235 Honey Mews
Caguas	Puerto Rico	Mary	2813	Smith	Consumer	PR	2585 Silent Autumn Landing
Caguas	Puerto Rico	Mary	11426	Smith	Consumer	PR	8663 Crystal Bluff Walk
Caguas	Puerto Rico	Mary	11664	Smith	Consumer	PR	3180 Golden Cider Circle
Caguas	Puerto Rico	Mary	5243	Smith	Consumer	PR	2387 Silver Freeway
Caguas	Puerto Rico	Mary	2214	Smith	Consumer	PR	6567 Red Sky Gate
Caguas	Puerto Rico	Mary	9072	Smith	Consumer	PR	4948 Stony Close
Caguas	Puerto Rico	Mary	5038	Smith	Consumer	PR	6847 Broad Orchard
Caguas	Puerto Rico	Mary	7101	Smith	Consumer	PR	9503 Hazy Pine Edge
Caguas	Puerto Rico	Mary	2402	Smith	Consumer	PR	6759 Clear Maze
Caguas	Puerto Rico	Mary	10517	Smith	Consumer	PR	3261 Heather Estates
Caguas	Puerto Rico	Mary	7270	Smith	Consumer	PR	100 Clear Forest Highway
Caguas	Puerto Rico	Mary	1507	Smith	Consumer	PR	436 Noble Shadow Estates
Caguas	Puerto Rico	Mary	6247	Smith	Consumer	PR	1070 Wishing Panda Ramp
Caguas	Puerto Rico	Mary	577	Smith	Consumer	PR	6517 Hazy Wagon Walk
Caguas	Puerto Rico	Mary	2588	Smith	Consumer	PR	1497 Little Walk
Caguas	Puerto Rico	Mary	9264	Smith	Consumer	PR	4204 Iron Hollow
Caguas	Puerto Rico	Mary	635	Smith	Consumer	PR	8621 Broad Forest
Caguas	Puerto Rico	Mary	4474	Smith	Consumer	PR	8659 Foggy Deer Port
Caguas	Puerto Rico	Mary	800	Smith	Consumer	PR	3450 Quiet Hickory Harbour

Dim_payment:

Type ▾	Late_delivery_risk ▾	Customer City ▾	Customer Country ▾	order date (DateOrders) ▾	Order Id ▾	Order Profit Per Order ▾	Order Region ▾	Product Name ▾
DEBIT	1	Caguas	Puerto Rico	9/20/2015 21:21	18010	58.5	Western Europe	Nike Men's CJ Elite 2 TD Football Cleat
DEBIT	0	Caguas	Puerto Rico	9/1/2015 5:26	16663	-107.1999969	Western Europe	Nike Men's CJ Elite 2 TD Football Cleat
DEBIT	1	Caguas	Puerto Rico	8/2/2015 2:40	14600	34.75	Western Europe	Nike Men's CJ Elite 2 TD Football Cleat
DEBIT	1	Caguas	Puerto Rico	7/18/2015 3:34	13575	12.86999989	Western Europe	Nike Men's CJ Elite 2 TD Football Cleat
DEBIT	1	Caguas	Puerto Rico	7/12/2015 12:41	13190	-157.6499939	Western Europe	Nike Men's CJ Elite 2 TD Football Cleat
DEBIT	1	Caguas	Puerto Rico	10/18/2015 9:24	19894	38.22000122	Western Europe	Nike Men's CJ Elite 2 TD Football Cleat
DEBIT	1	Caguas	Puerto Rico	9/6/2015 5:15	17005	7.639999866	Western Europe	Nike Men's CJ Elite 2 TD Football Cleat
DEBIT	0	Caguas	Puerto Rico	8/10/2017 5:40	65233	-17.3999962	Western Europe	Nike Men's CJ Elite 2 TD Football Cleat
DEBIT	0	Caguas	Puerto Rico	8/6/2017 12:20	64978	6.300000191	Western Europe	Nike Men's CJ Elite 2 TD Football Cleat
DEBIT	0	Caguas	Puerto Rico	6/21/2017 13:48	61831	-92.41999817	Western Europe	Nike Men's CJ Elite 2 TD Football Cleat
DEBIT	0	Caguas	Puerto Rico	8/31/2016 22:58	41717	44.13000107	Western Europe	Nike Men's CJ Elite 2 TD Football Cleat
DEBIT	1	Caguas	Puerto Rico	9/13/2015 19:26	17525	60.52000046	Western Europe	Nike Men's CJ Elite 2 TD Football Cleat
DEBIT	0	Caguas	Puerto Rico	7/5/2015 15:19	12718	34.04000092	Western Europe	Nike Men's CJ Elite 2 TD Football Cleat
DEBIT	0	Caguas	Puerto Rico	9/20/2017 23:39	68093	40.56000137	Western Europe	Nike Men's CJ Elite 2 TD Football Cleat
DEBIT	1	Caguas	Puerto Rico	6/18/2017 4:52	61600	33.68999863	Western Europe	Nike Men's CJ Elite 2 TD Football Cleat
DEBIT	0	Caguas	Puerto Rico	8/31/2015 23:29	16646	46.79999924	Western Europe	Nike Men's CJ Elite 2 TD Football Cleat
DEBIT	1	Caguas	Puerto Rico	9/19/2017 8:04	67980	37.04999924	Western Europe	Nike Men's CJ Elite 2 TD Football Cleat
DEBIT	0	Caguas	Puerto Rico	9/19/2015 16:16	17927	59.27999878	Western Europe	Nike Men's CJ Elite 2 TD Football Cleat
DEBIT	0	Caguas	Puerto Rico	9/5/2015 9:17	16948	-82.37000275	Western Europe	Nike Men's CJ Elite 2 TD Football Cleat
DEBIT	1	Caguas	Puerto Rico	7/16/2015 10:56	13459	43.22000122	Western Europe	Nike Men's CJ Elite 2 TD Football Cleat
DEBIT	0	Caguas	Puerto Rico	6/28/2015 14:27	12236	32.11000061	Western Europe	Nike Men's CJ Elite 2 TD Football Cleat
DEBIT	0	Caguas	Puerto Rico	9/16/2017 17:21	67801	-21.62000084	Western Europe	Nike Men's CJ Elite 2 TD Football Cleat
DEBIT	0	Caguas	Puerto Rico	8/10/2017 21:05	65277	27.63999939	Western Europe	Nike Men's CJ Elite 2 TD Football Cleat
DEBIT	0	Caguas	Puerto Rico	8/9/2017 10:24	65178	-4.670000076	Western Europe	Nike Men's CJ Elite 2 TD Football Cleat
DEBIT	0	Caguas	Puerto Rico	6/20/2017 17:28	61773	31.94000053	Western Europe	Nike Men's CJ Elite 2 TD Football Cleat
DEBIT	1	Caguas	Puerto Rico	6/18/2017 4:31	61599	49.13999939	Western Europe	Nike Men's CJ Elite 2 TD Football Cleat
DEBIT	1	Caguas	Puerto Rico	10/20/2015 5:11	20019	-22.72999954	Western Europe	Nike Men's CJ Elite 2 TD Football Cleat
DEBIT	0	Caguas	Puerto Rico	9/11/2017 21:44	67471	28.14999962	Western Europe	Nike Men's CJ Elite 2 TD Football Cleat
DEBIT	0	Caguas	Puerto Rico	8/1/2015 20:43	14583	34.29999924	Western Europe	Nike Men's CJ Elite 2 TD Football Cleat
DEBIT	1	Caguas	Puerto Rico	7/16/2015 7:47	13450	34.29999924	Western Europe	Nike Men's CJ Elite 2 TD Football Cleat
DEBIT	1	Caguas	Puerto Rico	6/29/2015 21:17	12324	37.02999878	Western Europe	Nike Men's CJ Elite 2 TD Football Cleat

Dim_product details:

Type	Days for shipping (real)	Days for shipment (scheduled)	Delivery Status	Late_delivery_risk	Category Id	Category Name	Customer Zipcode	Department Id	Department Na
DEBIT	6	2	Late delivery	1	73	Sporting Goods	725	2	Fitness
DEBIT	1	0	Late delivery	1	73	Sporting Goods	725	2	Fitness
DEBIT	5	2	Late delivery	1	73	Sporting Goods	725	2	Fitness
DEBIT	6	2	Late delivery	1	73	Sporting Goods	725	2	Fitness
DEBIT	2	4	Advance shipping	0	73	Sporting Goods	725	2	Fitness
DEBIT	3	2	Late delivery	1	70	Men's Clothing	725	4	Apparel
DEBIT	5	2	Late delivery	1	76	Women's Clothing	725	4	Apparel
DEBIT	2	2	Shipping on time	0	76	Women's Clothing	725	4	Apparel
DEBIT	3	2	Late delivery	1	60	Baby	725	4	Apparel
DEBIT	4	2	Late delivery	1	75	Video Games	725	9	Discs Shop
DEBIT	5	2	Late delivery	1	71	Music	725	9	Discs Shop
DEBIT	5	4	Late delivery	1	72	Pet Supplies	725	11	Pet Shop
DEBIT	5	4	Late delivery	1	72	Pet Supplies	725	11	Pet Shop
DEBIT	3	4	Advance shipping	0	72	Pet Supplies	725	11	Pet Shop
DEBIT	5	4	Late delivery	1	69	Health and Beauty	725	12	Health and Beaut
DEBIT	3	4	Advance shipping	0	69	Health and Beauty	725	12	Health and Beaut
DEBIT	5	4	Late delivery	1	69	Health and Beauty	725	12	Health and Beaut
DEBIT	3	4	Advance shipping	0	69	Health and Beauty	725	12	Health and Beaut
DEBIT	4	4	Shipping on time	0	69	Health and Beauty	725	12	Health and Beaut
DEBIT	3	4	Advance shipping	0	76	Women's Clothing	725	4	Apparel
DEBIT	2	4	Advance shipping	0	76	Women's Clothing	725	4	Apparel
DEBIT	3	4	Advance shipping	0	76	Women's Clothing	725	4	Apparel
DEBIT	3	4	Advance shipping	0	66	Crafts	725	4	Apparel
DEBIT	5	4	Late delivery	1	66	Crafts	725	4	Apparel
DEBIT	4	4	Shipping on time	0	63	Children's Clothing	725	4	Apparel
DEBIT	2	4	Advance shipping	0	76	Women's Clothing	725	4	Apparel
DEBIT	2	4	Advance shipping	0	76	Women's Clothing	725	4	Apparel
DEBIT	2	4	Advance shipping	0	76	Women's Clothing	725	4	Apparel
DEBIT	4	4	Shipping on time	0	68	Garden	725	6	Outdoors
DEBIT	4	4	Shipping on time	0	68	Garden	725	6	Outdoors
DEBIT	2	4	Advance shipping	0	68	Garden	725	6	Outdoors

Fact_ DataCoSupplyChain:

Type	Days for shipping (real)	Days for shipment (scheduled)	Benefit per order	Sales per customer	Delivery Status	Late_delivery_risk	Category Id	Category Name
DEBIT	6	2	131.1699982	285.1400146	Late delivery	1	73	Sporting Goods
DEBIT	1	0	35	309.7200012	Late delivery	1	73	Sporting Goods
DEBIT	5	2	115.4000015	317.9200134	Late delivery	1	73	Sporting Goods
DEBIT	6	2	-7.86999886	314.6400146	Late delivery	1	73	Sporting Goods
DEBIT	2	4	29.82999992	298.25	Advance shipping	0	73	Sporting Goods
DEBIT	3	2	61.99000168	206.6300049	Late delivery	1	70	Men's Clothing
DEBIT	5	2	52.5699997	187.7599945	Late delivery	1	76	Women's Clothing
DEBIT	2	2	18.78000069	187.7599945	Shipping on time	0	76	Women's Clothing
DEBIT	3	2	24.61000061	50.22000122	Late delivery	1	60	Baby
DEBIT	4	2	10.34000015	39.75	Late delivery	1	75	Video Games
DEBIT	5	2	63.49000168	226.7700043	Late delivery	1	71	Music
DEBIT	5	4	19.62000084	78.48999786	Late delivery	1	72	Pet Supplies
DEBIT	5	4	30.84000015	73.43000031	Late delivery	1	72	Pet Supplies
DEBIT	3	4	32.91999817	70.05000305	Advance shipping	0	72	Pet Supplies
DEBIT	5	4	137.8500061	287.1799927	Late delivery	1	69	Health and Beauty
DEBIT	3	4	100.5199966	276.9200134	Advance shipping	0	69	Health and Beauty
DEBIT	5	4	-10.13000011	266.6700134	Late delivery	1	69	Health and Beauty
DEBIT	3	4	-24.02000046	257.8800049	Advance shipping	0	69	Health and Beauty
DEBIT	4	4	21.10000038	234.4299927	Shipping on time	0	69	Health and Beauty
DEBIT	3	4	37.77000046	215.8200073	Advance shipping	0	76	Women's Clothing
DEBIT	2	4	67.55000305	215.8200073	Advance shipping	0	76	Women's Clothing
DEBIT	3	4	72.22000122	213.6600037	Advance shipping	0	76	Women's Clothing
DEBIT	3	4	79.13999939	452.25	Advance shipping	0	66	Crafts
DEBIT	5	4	134.2899933	447.6400146	Late delivery	1	66	Crafts
DEBIT	4	4	-160.2899933	337.4599915	Shipping on time	0	63	Children's Clothing
DEBIT	2	4	-161.8000031	194.2400055	Advance shipping	0	76	Women's Clothing
DEBIT	2	4	61.02000046	187.7599945	Advance shipping	0	76	Women's Clothing
DEBIT	2	4	1.830000043	183.4499969	Advance shipping	0	76	Women's Clothing
DEBIT	4	4	163.3600006	521.9299927	Shipping on time	0	68	Garden
DEBIT	4	4	216.9700012	516.5999756	Shipping on time	0	68	Garden
DEBIT	2	4	114.93000013	442.0400085	Advance shipping	0	68	Garden

Data Cleaning:

Data cleaning ensures that the supply chain datasets are accurate, consistent, and reliable for analysis. Without cleaning, dashboards may show misleading KPIs due to duplicates, nulls, or inconsistent formats.

Import Raw Data

Load CSV files into Power BI

- ***Import DataCoSupplyChainDataset.csv***
- ***Import DescriptionDataCoSupplyChain.csv***
- ***Import tokenized_access_logs.csv***
- ***Verify column headers and data types***

Handle Missing Values

Identify and treat null or blank fields.

- ***Replace missing numeric values with averages or medians***
- ***Fill categorical blanks with 'Unknown'***
- ***Drop records only if >50% fields are missing***

Remove Duplicates

Ensure each order and customer record is unique.

- ***Check duplicates in Order Id and Customer Id***
- ***Retain the first valid record***
- ***Remove redundant rows***

Standardize Formats

Align dates, currencies, and text fields.

- ***Convert order date and shipping date to YYYY-MM-DD***
- ***Ensure currency values are numeric***
- ***Normalize country names (e.g., 'USCA' → 'USA & Canada')***

Validate Relationships

Confirm integrity between tables.

- ***Match Customer Id with Order Customer Id***
- ***Ensure Product Id aligns with Category Id***
- ***Cross-check Market vs Order Region***

Generate calculated columns for KPIs.

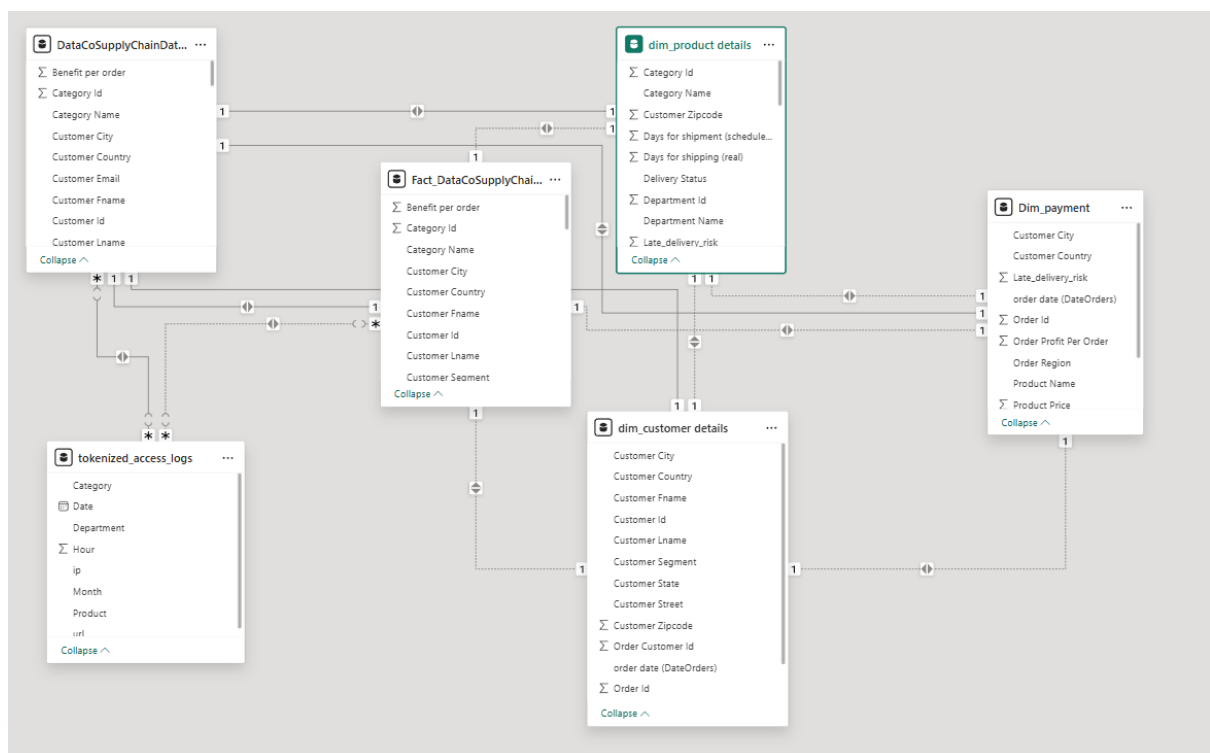
- ***Late Delivery Rate = Late orders ÷ Total orders***

- **Profit Margin % = Profit ÷ Sales × 100**
- **Customer LTV = $\Sigma(\text{Sales per Customer})$**

Data Modeling:

A star schema was implemented with **DataCoSupplyChainDataset** as the fact table and Date as dimensions.

Created one fact table and four dimensional table and create realtionships.



DAX Measures:

Ceated measures

Avg profit

total sales

total profit

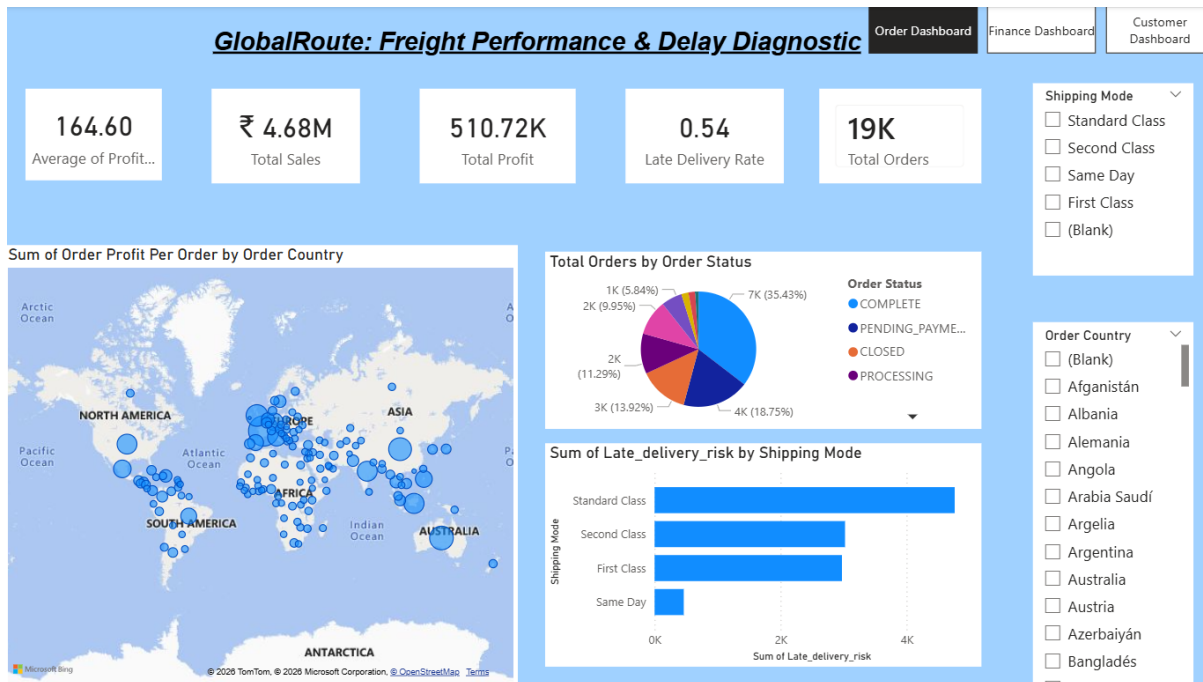
late delivery rate

total orders
on_time delivery
avg shipping dealy
profit margin
total revenue
avg order value
total discount
coustomer LTV
top products
pending orders

3. Dashboard Design:

Page 1:Order Diagnostics:

- **KPIs: Avg Order Value (₹226.44), Customer LTV (₹4.68M), Pending Orders (1889).**
- **Donut chart: Total Sales by Month (equal distribution across Sep–Jan).**
- **Market table: Freight performance by region (Africa, Europe, LATAM, Pacific Asia, USCA).**
- **Bar chart: Order Status Distribution (Complete, Pending, Processing, Fraud, etc.).**

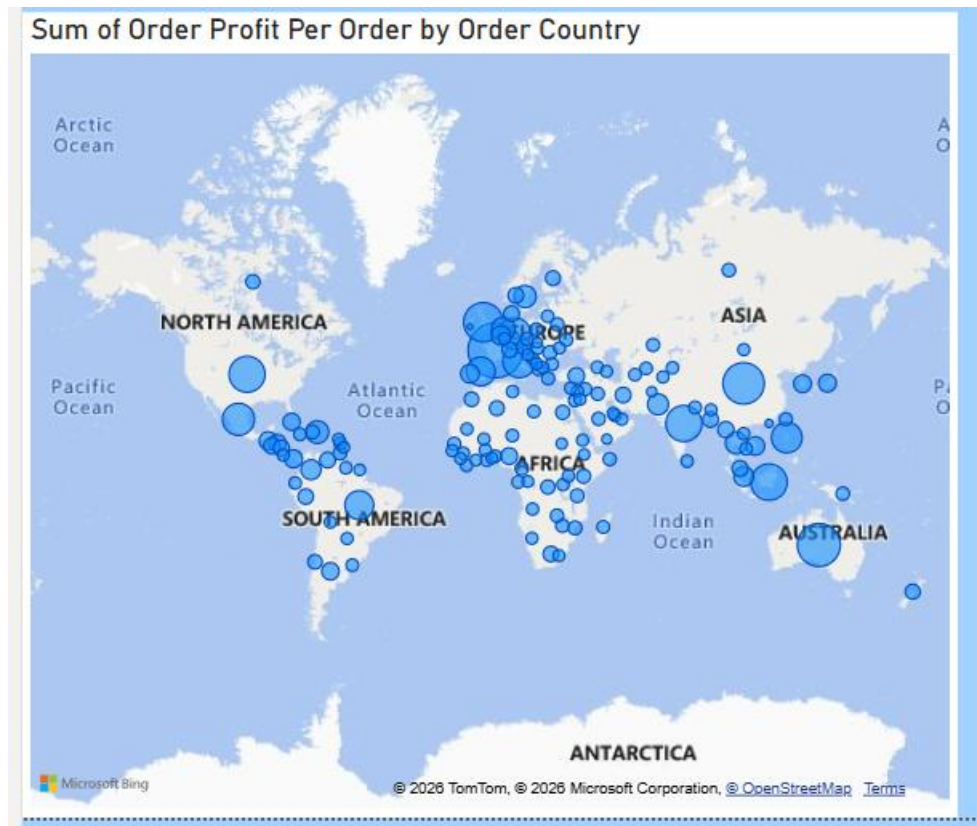


This page show which shipping modes cause the most delays?

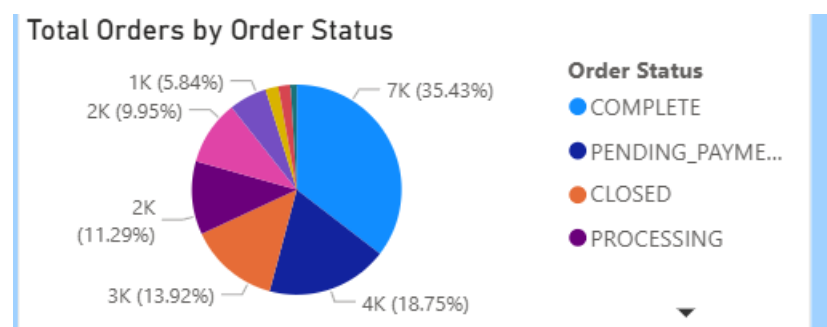
- ***List each KPI shown at the top Total Sales, Profit Margin %, Late Delivery Rate***
- ***For each metric:***
 - ***Define it clearly***
 - ***Late Delivery Rate indicates reliability of logistics operations***
 - ***A rate of 0.54 means 54% of orders are delayed, which is critical***

Visuals & Charts:

- **Map Visualization**
 - Describe what it shows (profits by country).
 - Explain how to interpret circle sizes/colors.
 - Connect to business insight (e.g., “High profits in Europe suggest strong demand, but delivery delays in LATAM reduce efficiency”).



- **Pie Chart (Orders by Status)**
 - List categories (Complete, Pending, Processing, etc.).
 - Explain proportions and what they mean for workflow health.
 - Highlight risks (e.g., “High Pending Payment orders may indicate billing issues”).



Bar Chart (Late Delivery Risk by Shipping Mode)

- Compare shipping modes.
- Note which mode has highest risk (Standard Class).
- Suggest actions (e.g., “Shift high-value customers to First Class to reduce risk”).



Filters

- Document available filters (Shipping Mode, Order Country).
- Explain how filters allow **drill-down analysis** (always check consistency when applying multiple filters).

Shipping Mode

☐ Standard Class

☐ Second Class

☐ Same Day

☐ First Class

☐ (Blank)

Order Country

☐ (Blank)

☐ Afganistán

☐ Albania

☐ Alemania

☐ Angola

☐ Arabia Saudí

☐ Argelia

☐ Argentina

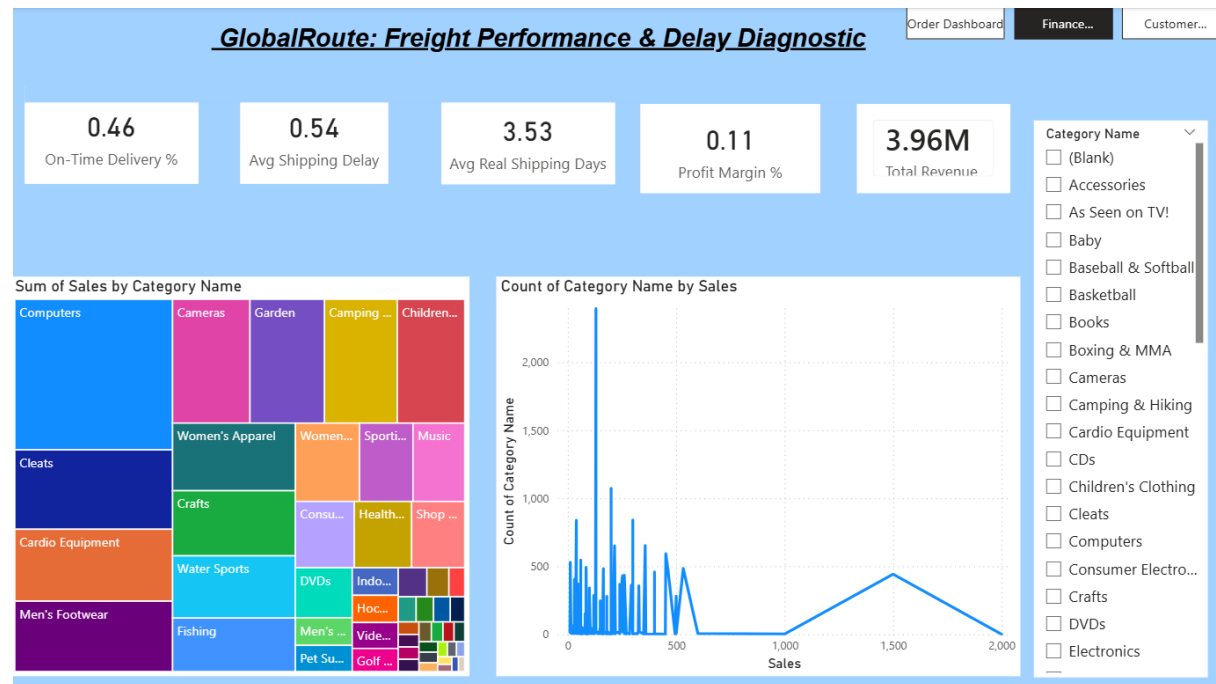
☐ Australia

☐ Austria

☐ Azerbaiyán

☐ Bangladés

Finance dashboard:



2. Key Metrics (Top Section)

- **On-Time Delivery % (0.46)**
 - Definition: Percentage of orders delivered on or before scheduled date.
 - Interpretation: Only 46% of deliveries are on time, indicating significant logistics inefficiencies.
 - Business Impact: High risk of customer dissatisfaction and potential churn.
- **Avg Shipping Delay (0.54)**
 - Definition: Average delay in shipping beyond scheduled days.
 - Interpretation: More than half a day's delay per order on average.
 - Business Impact: Suggests bottlenecks in Standard Class shipping.
- **Avg Real Shipping Days (3.53)**
 - Definition: Actual average time taken for delivery.
 - Interpretation: Customers wait ~3.5 days for products.
 - Business Impact: Benchmark against competitors nee
- **Profit Margin % (0.11)**
 - Definition: $\text{Profit} \div \text{Sales} \times 100$.
 - Interpretation: 11% margin is relatively low.
 - Business Impact: Discounts and freight costs may be eroding profitability.
- **Total Revenue (₹3.96M)**

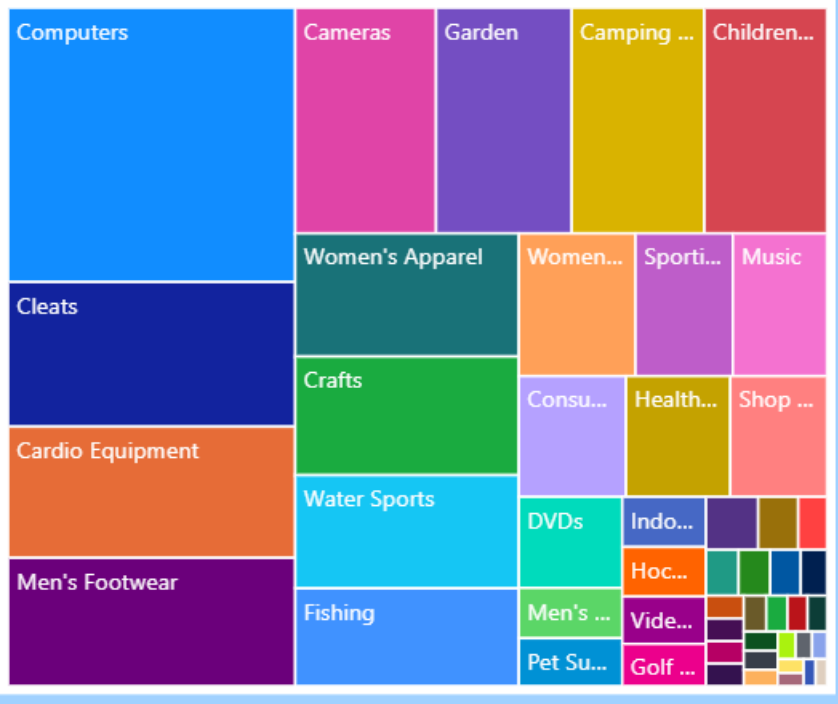
- Definition: Sum of all sales across categories.
- Interpretation: Strong revenue base, but margins need improvement.

Visuals & Charts:

Treemap – *Sum of Sales by Category Name*

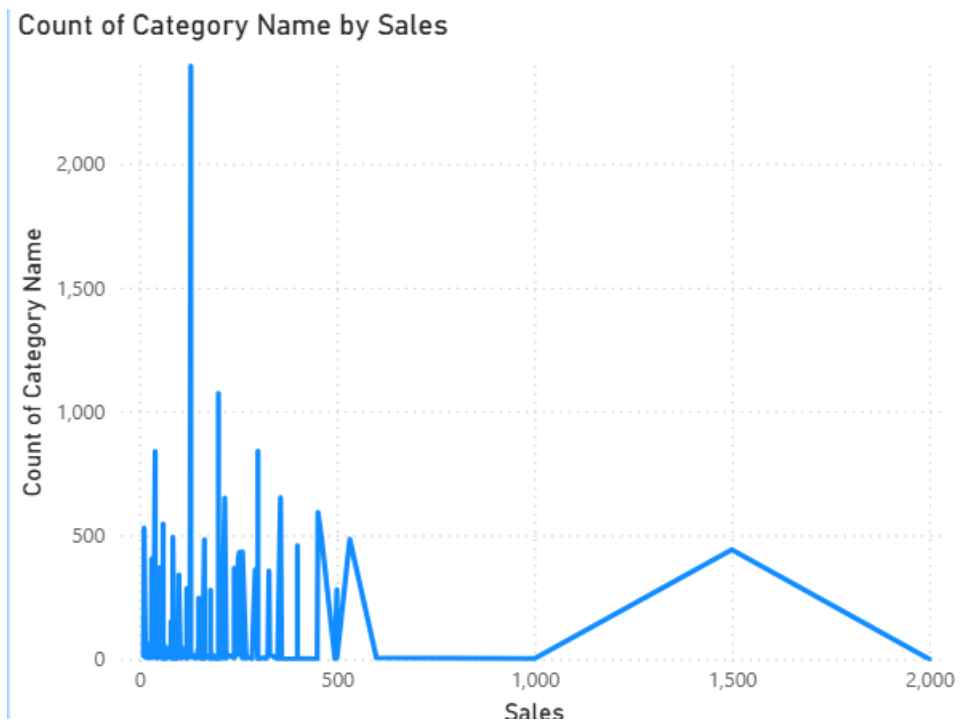
- Shows relative contribution of categories (Computers, Cameras, Apparel, Sports Equipment, etc.).
- Larger blocks = higher sales volume.
- Insight: Computers and Cameras dominate sales, while niche categories (Books, Crafts, Apparel) contribute less.
- Business Use: Helps prioritize inventory and marketing spend.

Sum of Sales by Category Name



Line Chart – *Count of Category Name by Sales*

- Plots sales counts across categories.
- Peak around 2,000 sales indicates top-selling categories.
- Insight: Certain categories consistently outperform others.
- Business Use: Identifies demand trends for forecasting.



Filters

- **Category Name Filter:** Allows drill-down into specific product categories (e.g., Electronics vs Apparel).
- **Best Practice:** Use filters to isolate underperforming categories and compare against high-performing ones.
- **Example:** Selecting “Computers” shows its dominance; selecting “Books” highlights niche contribution.

Category Name  

☐

(Blank)

☐

Accessories

☐

As Seen on TV!

☐

Baby

☐

Baseball & Softball

☐

Basketball

☐

Books

☐

Boxing & MMA

☐

Cameras

☐

Camping & Hiking

☐

Cardio Equipment

☐

CDs

☐

Children's Clothing

☐

Cleats

☐

Computers

☐

Consumer Electro...

☐

Crafts

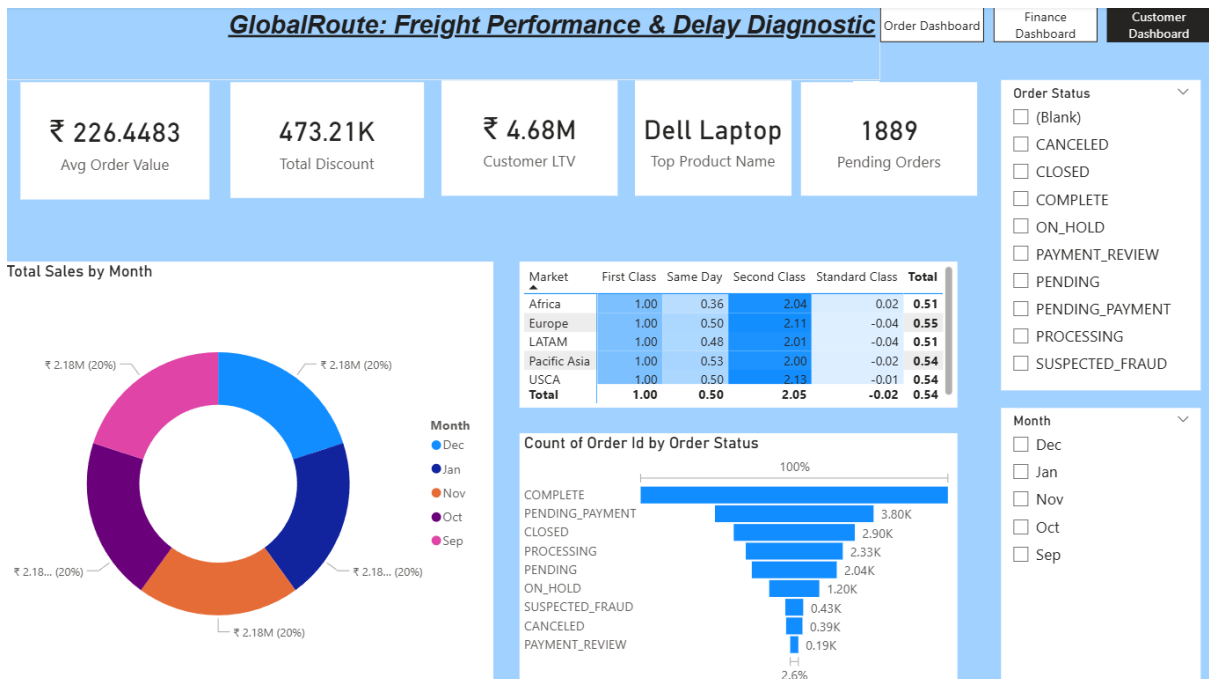
☐

DVDs

☐

Electronics

Customer Dashboard:



Month

☐

Dec

☐

Jan

☐

Nov

☐

Oct

☐

Sep

Total Sales by Month

₹ 2.18M (20%)

₹ 2.18M (20%)

₹ 2.18M (20%)

₹ 2.18M (20%)

₹ 2.18M (20%)

₹ 2.18M (20%)

Month

Dec

Jan

Nov

Oct

Sep

Market

First Class

Same Day

Second Class

Standard Class

Total

Africa

1.00

0.36

2.04

0.02

0.51

Europe

1.00

0.50

2.11

-0.04

0.55

LATAM

1.00

0.48

2.01

-0.04

0.51

Pacific Asia

1.00

0.53

2.00

-0.02

0.54

USCA

1.00

0.50

2.13

-0.01

0.54

Total

1.00

0.50

2.05

-0.02

0.54

Count of Order Id by Order Status

100%

COMPLETE

3.80K

PENDING_PAYMENT

2.90K

CLOSED

2.33K

PROCESSING

2.04K

PENDING

1.20K

ON_HOLD

0.43K

SUSPECTED_FRAUD

0.39K

CANCELED

0.19K

PAYMENT_REVIEW

2.6%

Key Metrics (Top Section)

- **Avg Order Value (₹226.44)**
 - Definition: Average revenue per order.
 - Insight: Indicates customer spending behavior.
 - Business Use: Helps forecast revenue per transaction.
- **Total Discount (473.21K)**
 - Definition: Sum of discounts applied across orders.
 - Insight: High discounts may erode profitability.
 - Business Use: Monitor promotional strategy effectiveness.
- **Customer LTV (₹4.68M)**
 - Definition: Lifetime value of customers.
 - Insight: Strong long-term revenue potential.
 - Business Use: Guides retention and loyalty programs.

Top Product Name (Dell Laptop)

- Definition: Highest-selling product.
- Insight: Dell Laptop drives significant sales.
- Business Use: Prioritize inventory and marketing for top products.

Pending Orders (1889)

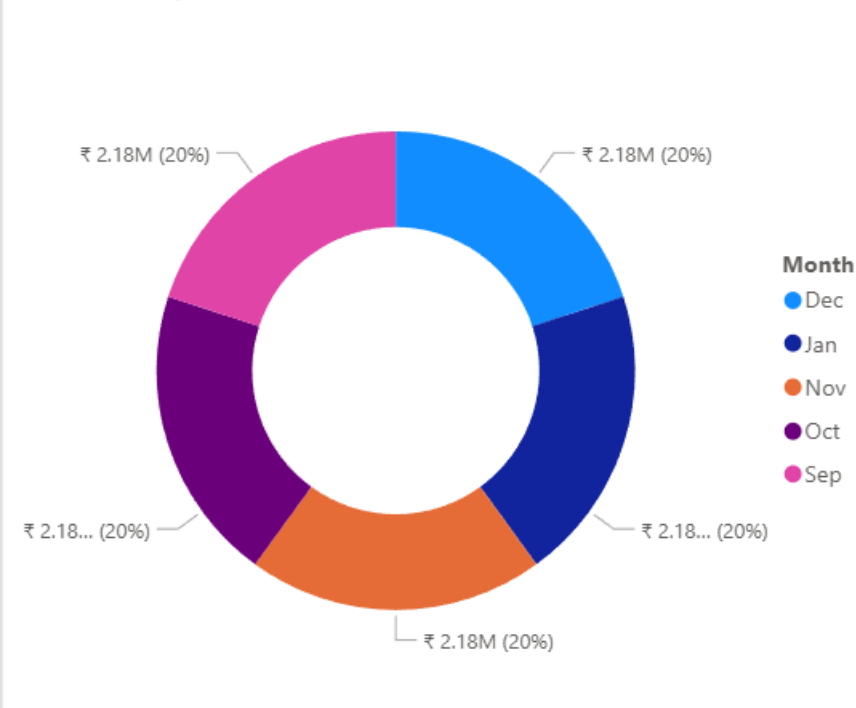
- Definition: Orders awaiting completion.
- Insight: High pending count signals workflow bottlenecks.
- Business Use: Improve order processing speed.

Visuals & Charts

Donut Chart – *Total Sales by Month*

- Shows equal distribution across Sep–Jan (₹2.18M each, 20%).
- Insight: Sales are stable across months, no seasonal spikes.
- Business Use: Predictable revenue stream, but opportunities exist to boost peak season sales.

Total Sales by Month



Market vs Shipping Class Table

- Compares freight performance across regions (Africa, Europe, LATAM, Pacific Asia, USCA).
- Shipping class totals:
 - First Class: 1.00
 - Same Day: 0.50
 - Second Class: 2.05
 - Standard Class: -0.02
 - Total: 0.54
- Insight: Standard Class shows negative performance, Second Class has highest delays.
- Business Use: Optimize shipping contracts and encourage premium modes for critical orders.

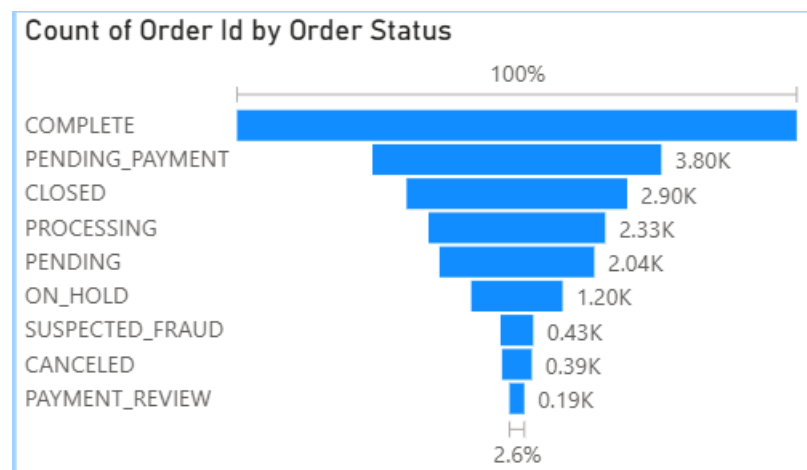
Market	First Class	Same Day	Second Class	Standard Class	Total
Africa	1.00	0.36	2.04	0.02	0.51
Europe	1.00	0.50	2.11	-0.04	0.55
LATAM	1.00	0.48	2.01	-0.04	0.51
Pacific Asia	1.00	0.53	2.00	-0.02	0.54
USCA	1.00	0.50	2.13	-0.01	0.54
Total	1.00	0.50	2.05	-0.02	0.54

Funnel Chart – *Count of Order Id by Order Status*

COMPLETE: 100%
PENDING_PAYMENT: 3.80K
CLOSED: 2.90K
PROCESSING: 2.33K
PENDING: 2.04K
ON_HOLD: 1.20K
SUSPECTED_FRAUD: 0.43K
CANCELED: 0.39K
PAYMENT_REVIEW: 0.19K

Insight: Large pending and processing volumes indicate operational inefficiencies. Fraud and canceled orders are relatively low but require monitoring.

Business Use: Streamline payment and processing workflows to reduce pending orders.



Filters

- **Order Status Filter:** Allows drill-down into specific statuses (e.g., Pending vs Complete).
- **Month Filter:** Enables analysis of monthly trends.
Use filters together to isolate problem areas (e.g., Pending Orders in November).

Order Status

- ☐ (Blank)
- ☐ CANCELED
- ☐ CLOSED
- ☐ COMPLETE
- ☐ ON_HOLD
- ☐ PAYMENT_REVIEW
- ☐ PENDING
- ☐ PENDING_PAYMENT
- ☐ PROCESSING
- ☐ SUSPECTED_FRAUD

Month

- ☐ Dec
- ☐ Jan
- ☐ Nov
- ☐ Oct
- ☐ Sep

Insights & Recommendations:

Freight Performance Insight

- **Late Delivery Rate (0.54):** More than half of orders are delayed, especially in *Standard Class*.
- **Regional Freight Trends:** Europe and USCA show relatively stable performance, while LATAM and Africa face higher delays.
- **Shipping Mode Comparison:** Standard Class consistently underperforms, while First Class and Same Day are more reliable.

Recommendations:

- Negotiate better contracts with Standard Class carriers.
- Incentivize customers to choose faster shipping modes for high-value orders.
- Implement predictive alerts for regions with recurring delays.

Customer Value Insights

- **Customer LTV (₹4.68M):** Indicates strong long-term revenue potential.
- **Avg Order Value (₹226.44):** Stable but could be increased with upselling strategies.
- **Pending Orders (1889):** High backlog risks customer dissatisfaction.

Recommendations:

- **Launch loyalty programs** to retain high-LTV customers.
- **Introduce cross-selling bundles** to raise average order value.
- **Automate order processing** to reduce pending orders.

. Financial Insights

- **Profit Margin (11%):** Relatively low compared to revenue.
- **Total Discounts (473K):** Discounts are eroding profitability.
- **Revenue Distribution:** Sales are evenly spread across months, showing no seasonal spikes.

Recommendations:

- **Rebalance discount strategy** to protect margins.
- **Focus on premium product categories** (Computers, Cameras) to sustain revenue.
- **Explore seasonal promotions** to create demand peaks.

Order Workflow Insights

- **Order Status Distribution:** Large volumes in *Pending Payment* and *Processing*.
- **Fraud & Cancellations:** Low but present (Suspected Fraud: 0.43K, Canceled: 0.39K).

Recommendations:

- **Strengthen payment gateway integration** to reduce pending payments.
- **Implement fraud detection algorithms** to flag high-risk orders.
- **Streamline approval processes** to reduce processing delays.

Product Category Insights

- **Top Product (Dell Laptop):** Drives significant sales.
- **Category Sales:** Computers and Cameras dominate, while niche categories (Books, Apparel) contribute less.

Recommendations:

- **Prioritize inventory for top-performing categories.**
- **Target marketing campaigns for niche categories to expand market share.**
- **Use treemap insights to identify underperforming categories for product strategy adjustments.**

Strategic Recommendations

- **Global Expansion: Focus on regions with strong profitability (Europe, USCA).**
- **Operational Improvements: Reduce shipping delays through logistics optimization.**
- **Future Enhancements: Integrate predictive analytics for delivery delays and ERP systems for real-time monitoring.**